

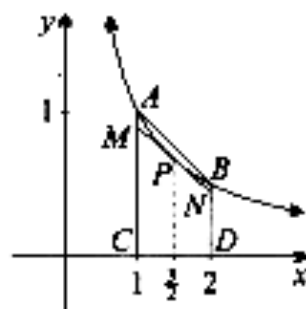
Inequalities WS 6

1. a. Prove by induction that $2^n > n$, for all positive integers n .
 b. Hence show that $1 < \sqrt[n]{n} < 2$, if n is a positive integer greater than 1.
 c. Suppose that a and n are positive integers. It is known that if $\sqrt[n]{a}$ is a rational number, then it is an integer. Explain why $\sqrt[n]{n}$, where n is a positive integer greater than 1, is never a rational number.
2. a. Show, using calculus, that the graph of $y = \ln x$ is concave down throughout its domain.
 b. Sketch the graph of $y = \ln x$, and mark two points $A(a, \ln a)$ and $B(b, \ln b)$ on the curve, where $0 < a < b$.
 c. Find the coordinates of the point P that divides the interval AB in the ratio 2:1.
 d. Using parts (b) and (c), deduce that $\frac{1}{3} \ln a + \frac{2}{3} \ln b < \ln \left(\frac{1}{3}a + \frac{2}{3}b \right)$.
3. a. If $a > 0, b > 0$ and $c > 0$ show that $(a + b)(b + c)(c + a) \geq 8abc$
 b. If $a > 0, b > 0, c > 0$ and $d > 0$ show that $(b + c + d)(a + c + d)(a + b + d)(a + b + c) \geq 81abcd$
4. a. If $a > 0, b > 0$ and $c > 0$, show that $a + b \geq 2\sqrt{ab}$
 b. Hence show that:
 - i. If $a > 0, b > 0$ and $c > 0$, then $(a + b)(b + c)(c + a) \geq 8abc$
 - ii. If $a > 0, b > 0, c > 0$ and $d > 0$, then $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4$
5. Given that $a + b + c = 1$ and $a + b + c \geq 3\sqrt[3]{abc}$ where a, b and c are positive real numbers:
 - a. Prove that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$
 - b. Hence or otherwise find the minimum value of $\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right)$
6. a. Prove that for positive numbers x and y that $x + y \geq 2\sqrt{xy}$
 b. Hence, prove $x^2 + \frac{1}{x^2} \geq 2$, for all real $x, x \neq 0$. Also, state for what value(s) of x there is equality.
 c. Hence, prove for positive numbers x, y and z that $(x + y)(y + z)(z + x) \geq 8xyz$
 d. By using part c, or otherwise, if a, b and c are the sides of a triangle, prove that

$$abc \geq (a + b - c)(b + c - a)(c + a - b)$$
7. a. Sketch the function $f(x) = \frac{1}{x}$
 b. Hence, prove $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n-1} \geq \log_e n$
8. Sketch the function $f(x)$ and prove $f(x) = \frac{1}{1+x^2}$: Subdivide the interval $[0, 1]$ into n equal parts. By considering the values of $f(x)$ at $x = 0, \frac{1}{n}, \frac{2}{n}, \dots, \frac{r}{n}, \dots, \frac{n}{n}$, prove that: Hint: $\left(\frac{n^2}{n^2 + r^2} = \frac{1}{1 + \left(\frac{r}{n}\right)^2} \right)$

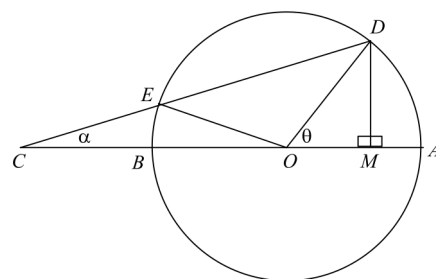
$$\frac{1}{n} \sum_{r=1}^n \frac{n^2}{n^2 + r^2} \leq \frac{\pi}{4}$$

9. The points A, P and B on the curve $y = \frac{1}{x}$ have x-coordinates $1, 1\frac{1}{2}$ and 2 respectively. The points C and D are the feet of the perpendiculars drawn from A and B to the x-axis. The tangent to the curve at P cuts AC and BD at M and N respectively.

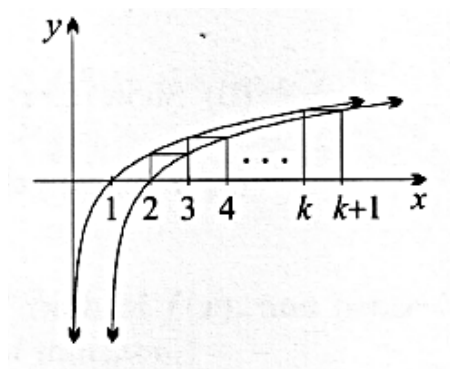


- Find the area of trapezia ABDC and MNDC.
 - Hence show that $\frac{2}{3} < \ln 2 < \frac{3}{4}$.
10. Consider the function $f(x) = \frac{(n+1+x)^{n+1}}{(n+x)^n}$, $x \geq 0$ where $n \geq 1$ is a fixed positive integer.
- Show that for $x > 0$, $f(x)$ is an increasing function of x
 - Hence, show that $\left(1 + \frac{x}{n+1}\right)^{n+1} > \left(1 + \frac{x}{n}\right)^n$
 - Deduce that $(n+2)^{n+1}n^n > (n+1)^{2n+1}$

11. In the diagram AB is a diameter of a circle with centre O and radius 1. C is a point on AB produced such that $BC = AO = OB$. D is a point on the circle such that $\angle AOD = \theta$, $0 < \theta < \frac{\pi}{2}$. CD cuts the circle at E and $\angle BCE = \alpha$. M is the foot of the perpendicular from D to AB.



- Show that $\tan \alpha = \frac{\sin \theta}{2 + \cos \theta}$
 - Explain why $\angle BOE = \alpha + \epsilon$ for some $\epsilon > 0$. Hence show that $\theta = 3\alpha + \epsilon$
 - Hence show that $\frac{\sin \theta}{2 + \cos \theta} < \tan \frac{\theta}{3}$ for $0 < \theta < \frac{\pi}{2}$
12. The diagram shows the curves $y = \log x$ and $y = \log(x-1)$, and $k-1$ rectangles constructed between $x=2$ and $x=k+1$, where $k \geq 2$.



- Show that:
 - $\int_2^{k+1} \log(x-1) dx = k \log k - k + 1$
 - $\int_2^{k+1} \log x dx = (k+1) \log(k+1) - \log 4 - k + 1$
- Deduce that $k^k < k! e^{k-1} < \frac{1}{4} (k+1)^{k+1}$, for all $k \geq 2$.

Answer

- c. $\left(\frac{a+2b}{3}, \frac{\ln a + 2 \ln b}{3}\right)$
- b. 9
- b. Equality is when $x = \pm 1$
 $\frac{3}{4}$ and $\frac{2}{3}$ square units.